

```

#include <stdio.h>

#define MAX 5

int queue[MAX];

int front = -1, rear = -1;

void insert(int value) {
    if (rear == MAX - 1) {
        printf("Queue Overflow! Cannot insert %d\n", value);
    } else {
        if (front == -1) front = 0;
        queue[++rear] = value;
        printf("%d inserted into queue.\n", value);
    }
}

void delete() {
    if (front == -1 || front > rear) {
        printf("Queue Underflow! Cannot delete.\n");
    } else {
        printf("%d deleted from queue.\n", queue[front]);
        front++;
    }
}

void display() {
    if (front == -1 || front > rear) {
        printf("Queue is empty.\n");
    } else {
        printf("Queue elements: ");
        for (int i = front; i <= rear; i++) {
            printf("%d ", queue[i]);
        }
    }
}

```

```

        printf("\n");
    }
}

int main() {
    int choice, value;
    while (1) {
        printf("\n--- Queue Menu ---\n");
        printf("1. Insert\n2. Delete\n3. Display\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                insert(value);
                break;
            case 2:
                delete();
                break;
            case 3:
                display();
                break;
            case 4:
                printf("Exiting program.\n");
                return 0;
            default:
                printf("Invalid choice! Try again.\n"); }
        }
}

```

Output

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 1
1 inserted into queue.
```

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 2
2 inserted into queue.
```

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 3
3 inserted into queue.
```

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 2
1 deleted from queue.
```

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 3
Queue elements: 2 3
```

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 23
Invalid choice! Try again.
```

```
--- Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 4
Exiting program.
```

```
Process returned 0 (0x0)   execution time : 25.343 s
Press any key to continue.
```